Literature Review

Tractor PTO-Driven Wood Chipper / Branch Mulcher

Power Source: New Holland 4710 4WD Tractor PTO (540 RPM)

PEFNAA Engineering Project #003

1. Introduction

Agricultural waste management has become a critical challenge in developing and developed economies alike. The disposal of tree branches, pruning waste, coconut fronds, and other lignocellulosic biomass through open burning causes significant air pollution, carbon emissions, and public health hazards. A tractor power-take-off (PTO)-driven wood chipper provides an efficient mechanical solution by converting this organic waste into wood chips and mulch, which can be used as biomass fuel, compost material, or soil amendment.

PEFNAA Engineering Project #003 tasks the team with designing and re-engineering a tractor PTO-driven wood chipper / branch mulcher, with the power source defined as the New Holland 4710 4WD tractor operating at a standard PTO output of 540 RPM. The New Holland 4710 is a widely used 55 HP four-wheel-drive utility tractor common in South and Southeast Asian agricultural contexts, making it a highly appropriate reference platform for a machine intended for local manufacture and deployment in Sri Lankan plantation and farming operations (PEFNAA Project Brief, 2026).

The design objective is to produce a machine that can be manufactured locally at an affordable cost for small and medium-scale farmers, while incorporating engineering quality and safety standards sufficient to support future export potential. This literature review surveys the theoretical engineering foundations and commercial machine landscape necessary to define the design direction prior to commencing CAD modelling, simulation, and analytical calculations.

2. Wood Chipping Mechanisms and Cutting Theory

The fundamental operating principle of a wood chipper is the severing of wood fibres through a high-velocity cutting action between a rotating blade mounted on a disc or drum rotor and a fixed counter-knife. Three principal chipping mechanisms are employed in industrial and agricultural chippers: disc chippers, drum chippers, and screw (cone) chippers (Nati et al., 2010).

2.1 Disc-Type Chipping Mechanism

In a disc chipper, cutting blades are mounted radially on the face of a heavy rotating disc. As the disc rotates, branches fed through the infeed chute are sliced by the blade against a fixed bed knife. The disc itself acts as a flywheel, storing kinetic energy between cutting events and smoothing the torque demand on the power source. This design produces chips of relatively uniform length, determined by the feed rate and rotational speed, making it highly suitable for biomass fuel production and composting applications (Spinelli et al., 2011).

The chip length L produced by a disc chipper is governed by the relationship between feed velocity and disc kinematics:

Chip Length Formula

$$L = v_f / (n \times z)$$

where v_f is the material feed velocity (m/min), n is the rotational speed of the disc (rpm), and z is the number of blades on the disc. Optimising these parameters allows chip length to be controlled for specific end-use requirements (Hellström et al., 2011).

2.2 Drum-Type Chippers

Drum chippers mount cutting blades on the periphery of a cylindrical drum rotating about a horizontal axis. While drum chippers can process larger diameter material and exhibit more even power demand, they typically produce chips of less uniform geometry compared to disc designs. For agricultural applications requiring moderate chip uniformity, disc-type designs are generally preferred (Nati et al., 2010).

2.3 Cutting Force and Specific Energy

Understanding cutting forces is essential for rotor design, shaft sizing, and power estimation. The specific cutting energy E_s (J/m³) required to chip wood depends on wood species, moisture content, cutting angle, and blade sharpness. Studies indicate that specific cutting energy for fresh agricultural wood ranges from approximately 250 to 800 kJ/m³, with higher values observed for hardwoods and wet material (Spinelli and Visser, 2009). The instantaneous cutting force F_c acting on a blade is:

Cutting Force

$$F_c = E_s \times A_c$$

where A_c is the cross-sectional area of the chip being removed. Blade geometry — specifically the cutting angle (typically 30–45°) and clearance angle (5–10°) — significantly influences cutting efficiency and blade wear rate (Günter et al., 2012).

3. Power and Torque Transmission

The machine receives mechanical power from the New Holland 4710 tractor PTO shaft rotating at 540 RPM and transmits it to the chipping rotor through a series of mechanical elements including the PTO shaft, pulleys or gears, and the rotor shaft. The fundamental relationship between transmitted power and torque is (PEFNAA Project Brief, 2026):

Torque from Power

$$T = (9550 \times P) / N$$

where T is the torque in Newton-metres (Nm), P is the transmitted power in kilowatts (kW), and N is the rotational speed in RPM. As an example calculation using the project reference condition of 18 kW at 540 RPM:

Example: T at 540 RPM

$$T = (9550 \times 18) / 540 \approx 318 \text{ Nm}$$

This torque of approximately 318 Nm must be safely transmitted through every mechanical element in the drivetrain — PTO shaft, pulleys or gears, and the rotor shaft — without exceeding the yield or fatigue strength of any component. Each element must be sized accordingly, with appropriate safety factors applied to account for the impact and shock loading inherent in wood chipping operations.

3.1 PTO Shaft and Universal Joint Coupling

The drive connection between the tractor and chipper is made through a standard agricultural PTO shaft incorporating universal joints (Cardan joints) at each end, with a telescoping central section to accommodate variations in hitch geometry. The shaft is enclosed in a protective plastic shield in accordance with ISO 11684 and EN 9283 safety standards. PTO shaft torque rating must exceed the maximum torque under peak cutting conditions, with a minimum safety factor of 1.5–2.0.

3.2 Belt Drive Speed Step-Up

A V-belt drive steps the PTO input speed of 540 RPM up to the required rotor speed of approximately 900–1,800 RPM. The fundamental speed ratio relationship for a belt drive is (PEFNAA Project Brief, 2026):

Speed Ratio

$$N1 / N2 = D2 / D1$$

where N1 and N2 are the driving and driven shaft speeds (RPM) respectively, and D1 and D2 are the corresponding pulley diameters. For a target rotor speed of 1,620 RPM from a 540 RPM PTO input, a speed ratio of 3:1 is required, achievable with pulley diameters in a 1:3 ratio (e.g., 150 mm driver, 450 mm driven). The design power to be transmitted is amplified by a service factor K_s to account for shock loading:

Design Power

$$P_d = P_{rated} \times K_s$$

For heavy shock loads characteristic of wood chipping, $K_s = 1.4–1.8$ is recommended (Mott, 2014). V-belts are preferred over flat belts for this application due to their wedging action in the groove, which increases grip and reduces slip under shock loading.

4. Shaft Design — Torsion and Bending

The rotor shaft is subjected to combined torsional and bending loading. Torsion arises from the transmitted drive torque; bending arises from the weight of the rotor disc assembly acting over the shaft span and from cutting reaction forces. Both stress components must be evaluated and combined to determine the safe shaft diameter (PEFNAA Project Brief, 2026).

4.1 Torsional Shear Stress

The torsional shear stress in a solid circular shaft is:

Torsional Shear Stress

$$\tau = 16T / (\pi d^3)$$

where τ is the shear stress (MPa), T is the applied torque (Nm), and d is the shaft diameter (m). This expression shows that shear stress is highly sensitive to shaft diameter — reducing d by 20% increases τ by approximately 95%.

4.2 Bending Stress

The bending stress at the critical section of the shaft is:

Bending Stress

$$\sigma = 32M / (\pi d^3)$$

where M is the bending moment (Nm) at the critical cross-section, computed from the rotor disc weight and any asymmetric belt tension loading applied to the shaft.

4.3 Combined Stress — Maximum Shear Theory

Under simultaneous torsion and bending, the maximum shear stress theory (Tresca criterion) gives the governing combined stress:

Combined Maximum Shear Stress

$$\tau_{max} = \sqrt{[(\sigma/2)^2 + \tau^2]}$$

This combined stress must be less than the allowable shear stress of the shaft material (yield strength divided by the safety factor). For a rotor shaft, the recommended safety factor is 2.5–3, and the preferred shaft material is medium-carbon steel (AISI 1045 or equivalent, $S_y \approx 530$ MPa). The ASME shaft equation incorporating fatigue stress concentration factors K_f and K_{fs} provides a more rigorous diameter:

ASME Shaft Diameter

$$d^3 = (16/\pi S_y) \times \sqrt{[(K^i\ M^b)^2 + (K^{ks}\ T)^2]}$$

4.4 Bearing Selection

Rolling element bearings support the rotor shaft radial and axial loads. The equivalent dynamic bearing load is calculated as:

Equivalent Dynamic Load

$$P = X \cdot F_r + Y \cdot F_a$$

where F_r is the radial load, F_a is the axial (thrust) load, and X, Y are radial and axial load factors from the bearing manufacturer's catalogue. The bearing L10 life in millions of revolutions is:

Bearing L10 Life

$$L_{10} = (C / P)^3$$

where C is the dynamic load rating of the bearing (kN) and P is the equivalent dynamic load (kN). For agricultural equipment, a minimum L10 life of 5,000 operating hours is recommended. Spherical roller bearings are preferred for the rotor shaft application due to their tolerance of shaft misalignment and ability to carry both radial and moderate axial loads (Shigley et al., 2011).

5. Tensile and Shear Strength of Connections

Structural connections in the chipper — blade mounting bolts, shaft keys, and structural frame bolts — must resist both tensile and shear forces generated during cutting operations. The fundamental stress relationships are (PEFNAA Project Brief, 2026):

Tensile Stress

$$\sigma = F / A$$

Shear Stress

$$\tau = F / A$$

where F is the applied force and A is the relevant cross-sectional area — the full bolt cross-section for tensile loading, or the bolt shear plane area for shear loading. Blade mounting bolts are particularly critical because they experience high-magnitude impact loads during each cutting event. Impact loading factors of 2–3 are applied to the static force estimates when sizing blade mounting hardware. High-strength bolts of Grade 8.8 or higher are recommended for all blade attachment and rotor assembly connections, with the specified torque applied using a calibrated torque wrench to ensure correct preload.

6. Fracture Mechanics

Rotor blades and the shaft may develop micro-cracks over service life due to repeated impact loading against wood. Crack propagation under cyclic stresses can lead to sudden brittle fracture without prior visible deformation — a critical failure mode in high-speed rotating machinery. The stress intensity factor K quantifies the severity of the stress field at a crack tip (PEFNAA Project Brief, 2026):

Stress Intensity Factor

$$K = Y \sigma \sqrt{\pi a}$$

where K is the stress intensity factor (MPa√m), Y is a dimensionless geometry factor depending on crack and specimen geometry, σ is the applied stress (MPa), and a is the half crack length (m). Fracture occurs when K reaches or exceeds the material's fracture toughness:

Fracture Criterion

$$K \geq K_c$$

where K_c is the plane-strain fracture toughness of the blade or shaft material. For the cutting blades (D2 tool steel, HRC 58–62), fracture toughness values are typically in the range of 18–25 MPa√m. For the rotor shaft (AISI 1045 steel), K_c is approximately 50–60 MPa√m. Regular inspection and blade replacement intervals must be specified in the maintenance schedule to prevent crack propagation reaching the critical level.

7. Fatigue Failure Analysis

Rotating components — particularly the rotor shaft, blade mounting bolts, and belt drive pulleys — experience cyclic loading with each revolution. Repeated cycling of stresses below the static yield strength can initiate and propagate fatigue cracks, leading to eventual fracture at stress levels far below the material's ultimate tensile strength. Fatigue failure is the most common cause of unexpected mechanical failure in rotating agricultural machinery (Shigley et al., 2011).

7.1 Alternating Stress Amplitude

The alternating stress amplitude that governs fatigue damage is (PEFNAA Project Brief, 2026):

Alternating Stress

$$\sigma_a = (\sigma_{max} - \sigma_{min}) / 2$$

where σ_{max} and σ_{min} are the maximum and minimum stresses in each loading cycle. For a rotor shaft in fully reversed bending (common in rotating shafts), $\sigma_{min} = -\sigma_{max}$ and the alternating stress equals the bending stress amplitude.

7.2 Goodman Fatigue Criterion

The modified Goodman criterion accounts for the combined effect of alternating and mean stress on fatigue life:

Goodman Criterion

$$\sigma_a/S_e + \sigma_m/S_u \leq 1$$

where σ_a is the alternating stress amplitude, σ_m is the mean (static) stress, S_e is the endurance limit of the material (typically $0.5 \times S_u$ for steels), and S_u is the ultimate tensile strength. A Goodman ratio less than or equal to 1.0 indicates the component is expected to have infinite fatigue life under the given loading conditions. For the rotor shaft, Goodman analysis must account for stress concentration factors at keyways, shoulders, and thread roots, which are the most common sites for fatigue crack initiation.

8. Impact Energy and Rotor Flywheel Design

Cutting branches requires kinetic energy to be delivered rapidly at the instant of blade-wood contact. The chipper rotor disc functions as a flywheel, storing rotational kinetic energy that is released into the wood during each cutting stroke. The total kinetic energy stored in the rotating rotor is (PEFNAA Project Brief, 2026):

Kinetic Energy of Rotor

$$E = \tfrac{1}{2} \times I \times \omega^2$$

where I is the mass moment of inertia of the rotor disc assembly ($kg \cdot m^2$) and ω is the angular velocity of the disc (rad/s). For a solid disc of mass m and radius R, the moment of inertia is:

Moment of Inertia — Disc Rotor

$$I = \tfrac{1}{2} \times m \times r^2$$

A heavier rotor and/or larger radius both increase the stored kinetic energy, thereby improving cutting performance through larger and more irregular feed material. However, a heavier rotor also increases shaft bending load and requires a more powerful prime mover to spin up to operating speed. The design must therefore balance energy storage against shaft loading and tractor PTO power availability.

For example, a 100 kg disc of 0.45 m radius rotating at 1,620 RPM ($\omega = 169.6$ rad/s) stores:

Energy Storage Example

$$I = 0.5 \times 100 \times 0.45^2 = 10.125\ kg \cdot m^2 \quad \rightarrow \quad E = 0.5 \times 10.125 \times 169.6^2 \approx 145\ kJ$$

This is sufficient energy to chip branches up to 100–150 mm in diameter in a single pass, consistent with the 6-inch (150 mm) target capacity of the proposed design.

9. Vibration Analysis and Rotor Balancing

Rotating machines must avoid excessive vibration to prevent accelerated bearing wear, fastener loosening, and structural fatigue. Vibration in a wood chipper rotor system arises from three primary sources: rotor mass imbalance, uneven blade wear over time, and shaft or coupling misalignment (PEFNAA Project Brief, 2026).

9.1 Natural Frequency and Resonance Avoidance

The natural frequency of the rotor-shaft-bearing system must be established to ensure that the operating speed does not excite resonance. For a simplified model of a disc mounted on a shaft supported between bearings, the fundamental natural frequency is:

Natural Frequency

$$f_n = (1/2\pi) \times \sqrt{(k/m)}$$

where k is the effective bending stiffness of the shaft (N/m) and m is the mass of the rotor disc (kg). The critical speed N_c in RPM at which resonance occurs is related to the natural frequency by $N_c = 60 \times f_n$. Operating speed must be maintained at least 20–25% below the critical speed to provide an adequate safety margin against resonance excitation. For a typical chipper disc (100 kg, 900 mm shaft span), critical speeds are generally well above the 1,200–1,800 RPM operating range when standard shaft diameters of 50–65 mm are used.

9.2 Centrifugal Force from Imbalance

Even a small residual mass imbalance in the rotor creates a rotating centrifugal force that increases with the square of rotational speed:

Centrifugal Imbalance Force

$$F = m \times r \times \omega^2$$

where m is the imbalance mass (kg), r is its radial distance from the rotational axis (m), and ω is the angular velocity (rad/s). At operating speeds of 1,620 RPM ($\omega = 169.6$ rad/s), an imbalance of just 50 g at a radius of 200 mm generates a centrifugal force of approximately 288 N — sufficient to cause significant bearing loads and vibration. Dynamic balancing of the rotor disc to ISO 1940-1 Grade G6.3 is therefore required after blade installation and after any blade replacement.

10. Structural Frame Design and Stress Analysis

The structural frame of the chipper must support the rotor bearing housings, infeed chute, discharge chute, and three-point hitch attachment while withstanding dynamic loads generated during chipping. The frame is a welded steel structure and its critical members act as beams in bending. The bending stress in a beam cross-section is (PEFNAA Project Brief, 2026):

Bending Stress in Frame Member

$$\sigma = M \times y / I$$

where M is the bending moment at the section of interest (Nm), y is the perpendicular distance from the neutral axis to the outer fibre (m), and I is the second moment of area of the beam cross-section (m^4). For standard rectangular hollow sections (RHS) commonly available in Sri Lanka, I values are tabulated in structural steel section tables. The frame must be designed so that the maximum bending stress remains within the allowable stress, determined by dividing the yield strength of the frame steel (typically S355 or equivalent, $S_y = 355$ MPa) by the applicable safety factor.

Finite element analysis (FEA) provides a more comprehensive picture of stress distribution, particularly at welded joints and load attachment points. For a locally manufactured machine, analytical beam theory supplemented by engineering judgement at critical welded connections provides an acceptable first-pass design method, with FEA recommended for validation of the final design geometry.

11. Design Safety Factors

A safety factor (also called factor of safety, FS) is the ratio of the material strength to the applied stress in a component. It quantifies the margin of safety against failure and accounts for uncertainties in loading, material properties, manufacturing tolerances, and operating conditions (PEFNAA Project Brief, 2026):

Safety Factor Definition

$$FS = \text{Material Strength} / \text{Applied Stress}$$

For a PTO wood chipper operating under heavy shock loading conditions, the following safety factors are recommended by the project guidelines:

Component	Recommended Safety Factor
Rotor shaft	**2.5 – 3**
Structural frame	**2.0**
Bolts / fasteners	**3 – 4**
Cutting blades	**4.0**

These values reflect the higher criticality assigned to cutting blades and bolts — components that experience severe impact loads and whose sudden failure poses an immediate safety risk to the operator. Frame members are designed to a lower safety factor because structural steel frames provide visible warning of impending failure through plastic deformation, whereas blade and bolt fracture can be sudden and unpredictable. All safety factor calculations must be documented in the design analysis report and verified against the relevant mechanical engineering standards.

12. Feeding Systems

The feeding system governs the rate at which material is presented to the cutting disc and significantly affects throughput, operator safety, and power demand consistency. Two principal feeding system types are used in commercial PTO chippers.

12.1 Gravity and Manual Feed

In gravity-feed designs, material is loaded into an inclined infeed hopper and slides toward the cutting disc under gravity and manual pushing force. This approach is mechanically simple and low-cost, but is susceptible to jamming with wet, irregular, or leafy material. User experience data from the U.S. commercial market consistently identifies jamming as the primary operational limitation of gravity-feed designs.

12.2 Hydraulic Powered Infeed

Hydraulic infeed systems use a pair of driven feed rollers powered by a hydraulic motor to actively draw material into the cutting zone, powered from the tractor's external hydraulic system. Key advantages include automatic reversal on jam detection, consistent material feed rate regardless of branch orientation, and significantly improved operator safety. Industry analysis identifies hydraulic infeed as the defining feature of commercial and professional-grade machines (PEFNAA Project Brief, 2026).

The feed roller drive pressure is typically set to 100–200 bar, with a pressure relief valve triggering automatic roller reversal on jam detection. For a machine targeting professional-grade performance and commercial viability in the Sri Lankan market, hydraulic infeed is the recommended specification for the final design.

13. Commercial Machine Benchmarking

A review of the top commercial PTO-driven wood chippers available in the United States market, as compiled in PEFNAA Project Brief (2026), provides valuable benchmarks for key design parameters. Machines are categorised into three tiers based on capacity and feed system type.

13.1 Heavy / Professional Grade (6"–8", Hydraulic Infeed)

Machines in this category represent the commercial benchmark. All feature hydraulic infeed rollers, 6–7 inch chipping capacity, compatibility with 20–75 HP tractors, heavy flywheel disc, and multiple cutting blades.

Figure 1: Woodland Mills WC68 PTO Wood Chipper — USD 3,450 (6", hydraulic feed, 20–50 HP)

Figure 2: WoodMaxx MX-8600 PTO Wood Chipper — USD 4,790 (Made in USA, hydrostatic infeed)

Figure 3: BX72RSH 7" Hydraulic PTO Wood Chipper — USD 3,245 (contractor-grade, 7" capacity)

Figure 4: Farmry WCPH150 Hydraulic PTO Chipper — USD 2,999 (6" heavy-duty, 20–75 HP tractors)

Figure 5: MechMaxx WC68H Hydraulic PTO Chipper — USD 2,799 (commercial unit with hydraulic feeding)

The Woodland Mills WC68 is one of the most popular machines in North America and widely regarded as the benchmark value-for-money professional chipper. The WoodMaxx MX-8600, manufactured in the USA, incorporates hydrostatic infeed and targets the high-end commercial market. Farmry and MechMaxx offer competitive price points while retaining hydraulic infeed, making them the most directly relevant benchmarks for an affordable locally manufactured machine.

13.2 Mid-Range Semi-Commercial (5"–6")

This tier targets small farm and contractor applications. Machines typically use gravity feed and provide a useful price-point benchmark for a first-generation local design.

Figure 6: Titan Attachments PTO Wood Chipper 6" — USD 1,999 (popular for small contractors and farms)

Figure 7: WoodMaxx MX-8500G+ PTO Chipper — USD 2,990 (gravity-feed, low maintenance)

Figure 8: MechMaxx BX42S PTO Wood Chipper — USD 1,699 (entry-level commercial use)

Figure 9: NorTrac PTO Wood Chipper 5.5" — USD 1,799 (widely sold via farm equipment retailers)

Figure 10: Woodland Mills WC88 PTO Wood Chipper — USD 3,995 (8" capacity, strong market reputation)

13.3 Light / Small Farm Use (4"–5")

Entry-level machines in this tier handle branch diameters up to 4–5 inches with gravity feed and compact tractor compatibility. They provide the lowest price-point benchmarks for a locally manufactured entry-level model.

Figure 11: DR PTO Wood Chipper — USD 1,999 (handles ~4.75" branches, strong U.S. brand)

Figure 12: TMG Sub-Compact PTO Wood Chipper — USD 1,699 (designed for compact tractors)

Figure 13: Farmry 5" PTO Wood Chipper — USD 1,698 (budget-friendly entry-level design)

Figure 14: Bolton Tools 4" PTO Chipper — USD 1,344 (low-cost, hobby farm use)

Figure 15: Mighty Mac TPH185 PTO Chipper — USD 2,398 (U.S.-made hammermill style design)

13.4 Key Design Insights from Benchmarking

Across all tiers, the most competitive machines share: hydraulic infeed (professional grade), 6-inch chipping capacity, compatibility with 20–50 HP tractors, a heavy flywheel disc, two to four wear-resistant cutting blades, and a Category I/II three-point hitch. User feedback consistently identifies gravity-feed jamming as the primary limitation of lower-cost designs. The recommended winning formula for the proposed Sri Lankan machine is: 6-inch capacity, hydraulic infeed, heavy flywheel disc, robust blade and counter-knife design, and a fabrication-cost target achievable through local steel fabrication with imported blade, bearing, and hydraulic components.

14. Local Manufacturing and Market Context

Sri Lanka's agricultural sector generates large volumes of branch and pruning waste from coconut, rubber, and tea plantations annually. Current disposal methods — predominantly open burning — are environmentally harmful and represent a loss of potential biomass energy and soil amendment material. A locally designed and manufactured PTO wood chipper addresses this gap while providing an affordable alternative to imported machines, which face high import duties and limited local spare parts availability.

The PEFNAA project brief (2026) frames this project as the foundation of a potential product family encompassing wood chippers, animal feed choppers, silage cutters, and mulching machines — all sharing common engineering principles of cutting mechanism design, rotor dynamics, and power transmission. For local manufacturing viability, the design must use locally available structural steel sections and standard bearing, belt, and fastener sizes. Fabrication processes must be achievable in a well-equipped local workshop using cutting, bending, welding, and basic turning operations.

15. Summary and Research Gaps

This literature review has established the theoretical and practical engineering knowledge base necessary for the design of a tractor PTO-driven disc wood chipper powered by the New Holland 4710 at 540 RPM. The eleven key mechanical engineering concepts identified in the project brief — power and torque transmission, shaft torsion and bending, tensile and shear connection design, fracture mechanics, fatigue analysis, impact and kinetic energy, vibration and rotor balancing, bearing load and life, belt drive design, structural frame stress analysis, and safety factors — have each been addressed and contextualised within the commercial machine landscape. Key design conclusions are:

- Disc-type chipping with a high-inertia flywheel disc is the recommended mechanism for consistent chip quality and drivetrain protection.
- A V-belt speed step-up from 540 RPM to approximately 1,620 RPM, with K_s = 1.5–1.8, is the most practical transmission approach for local manufacture.
- Rotor shaft diameter must be determined by combined torsion-bending analysis using the ASME shaft equation with appropriate fatigue and stress concentration factors.
- Blade mounting hardware must be sized for impact load amplification factors of 2–3 and specified as Grade 8.8 or higher bolts with safety factors of 3–4.
- Fracture mechanics and fatigue analysis (Goodman criterion) are required for blades and shaft to establish inspection intervals and blade replacement schedules.
- Dynamic rotor balancing to ISO 1940-1 Grade G6.3 is mandatory after blade installation and replacement.
- Hydraulic infeed is recommended for the final design to meet professional-grade performance and commercial viability targets.
- All components must be designed with safety factors consistent with PEFNAA guidelines: shaft 2.5–3, frame 2.0, bolts 3–4, blades 4.0.

The primary research and design tasks for subsequent project phases are: cutting force measurement for locally occurring wood species (coconut, rubber, jak); optimisation of disc mass and blade geometry for 6-inch capacity; hydraulic circuit design and hydraulic motor sizing for the infeed system; structural frame FEA validation; and detailed manufacturing drawings and Bill of Materials for local fabrication.

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